Catching Silent Failures in LLM Agents

Beyond-Score Uncertainty from Diverse Critics — Research Proposal

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AT A GLANCE

Q. When an agent selects its next action (or trajectory) using a critic to score candidates, can the critic's *uncertainty* flag the "silent failures" — actions the critic rates highly but that actually derail the task?

How. Log an agent's candidate actions on verifiable tasks, take the ones it *selects*, and check whether uncertainty adds signal *beyond* the critic score itself — and whether it depends on the critics being genuinely *diverse*.

Test. ΔAUROC > 0 among the selected actions, with critic score and cheap features already controlled; and whether it rises with critic diversity.

Why it matters. A calibrated "this selected action is secretly failing" signal is exactly the missing trigger for backtracking / selective verification in reliable agents. A clean null is itself a safety-relevant result.

Working claim. An LLM agent commits to actions by taking the highest-scoring candidate under a critic (best-of-n, tree search, or a process reward model). Selecting the argmax of an imperfect critic concentrates on the critic's own mistakes — the *silent failures* that look good but derail the task. This project tests whether critic *uncertainty* carries information about those failures — conditional on selection, beyond the critic score itself — and whether that information is a property of *critic diversity* rather than of ensembling per se.

Research question. Among the actions (or trajectories) an LLM agent selects from a critic's top scores, does critic uncertainty discriminate task-verified failures from successes *beyond the critic score itself*, and how does this depend on *critic ensemble diversity*?

In plain terms. An agent's chosen next steps are a mix of real progress and moves that only *look* like progress to its critic. Among those chosen moves, does disagreement across a panel of critics reveal the silent failures the score alone missed — and only when the panel is genuinely different, not near-clones?

1 Background

- **Agents fail silently, mid-task, and detection is the bottleneck.** LLM agents loop, cascade tool errors, drift off-goal, or fabricate success; catching these before they compound is the reliability problem, and judging every step with a second LLM costs more than the agent itself (**AgentForesight**, arXiv:2605.08715; **Real-Time Detection and Repair of Agent Failures**, arXiv:2608.02464; survey, arXiv:2602.05073).
- **Agents commit by taking the argmax of a critic.** Best-of-n, tree search, and *process reward models* score candidate steps or trajectories and select the top one (**Let's Verify Step by Step**, arXiv:2305.20050; **Math-Shepherd**, arXiv:2312.08935; **Generative Verifiers**,

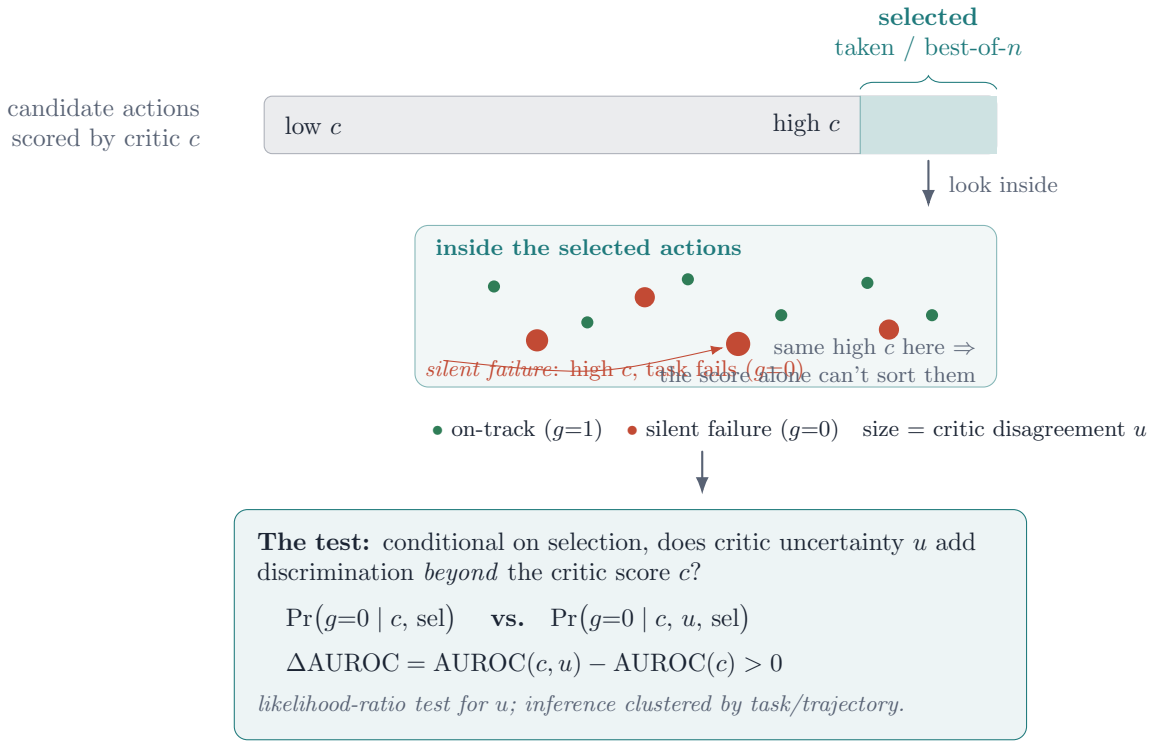


Figure 1: An agent's selected actions mix real progress with *silent failures* that overrated the critic. The estimand asks whether critic uncertainty u discriminates task-verified failures (g=0) from successes beyond the critic score c, conditional on selection ($\Delta\text{AUROC} > 0$).

arXiv:2408.15240; **AgentPRM**, arXiv:2511.08325). Selecting the maximum of an imperfect critic is exactly the operation that surfaces the critic's mistakes — the silent failures that look good to it.

- **Uncertainty tracks reliability — but only unconditionally, and outside agents.** Reward-model ensemble uncertainty filters low-quality outputs (**URM/URME**, arXiv:2410.00847; ensembles, arXiv:2310.02743; diversity, arXiv:2312.09244) and over-optimization worsens with pressure (arXiv:2210.10760) — but shown *marginally*, on single-answer non-agentic tasks, never conditioned on the critic score in the selected set.
- **Self-correction needs a trigger.** Reflexion-style verbal self-correction and state rollback can recover from a bad step *if they know when to fire* (**Reflexion**, arXiv:2303.11366). A calibrated "this selected action is secretly failing" signal is precisely that missing trigger.

2 Gap / Novelty

No published work asks the agentic, *selection-conditional, beyond-score* question: among the actions an agent *selects*, does critic *uncertainty* catch silent failures beyond the critic score, with critic *diversity manipulated and mediated*, and gold from *task success*. Prior agent work either scores steps (PRMs, generative verifiers), reflects/repairs after a known failure (Reflexion), or detects failures without asking whether uncertainty adds over the score in the selected set. The contribution is three tests: (i) *detection* — does critic uncertainty carry silent-failure information

beyond the score, conditional on selection; (ii) *mechanism* — does critic diversity govern it via error decorrelation ("more critics" vs "different critics"); (iii) *practice* — how much survives cheap uncertainty estimators, and what it buys as a backtrack / selective-verification trigger.

3 Hypotheses

H1 (beyond-score detection at selection). Among selected actions, critic uncertainty adds discrimination beyond the critic score (ΔAUROC > 0, LR test); the increment shrinks as selection pressure grows (larger n, greedier argmax), because selection concentrates on the critics' shared blind spots.

H2 (diversity, mediated). The increment rises as critics get more diverse (seed $\rightarrow$ data $\rightarrow$ base model $\rightarrow$ public-critic zoo), mediated by error decorrelation: a diversity step that fails to decorrelate critic errors fails to raise ΔAUROC.

H3 (pressure $\times$ diversity). The decay of ΔAUROC under stronger selection is flatter for base-diverse critic panels: diversity buys robustness to optimization pressure, not just level.

H4 (cheap uncertainty). Single-critic uncertainty (dropout, quantile-width, generative-verifier self-consistency) is tested against true critic ensembles on the same estimand; parity and gap are both headlines.

H5 (failure taxonomy). Detection is heterogeneous by failure class (tool/execution error, goal drift, hallucinated success, doom-loop repetition, verifier gaming), and per-class critic error correlation explains it: classes where diverse critics fail independently are the classes uncertainty catches.

Falsification is informative. ΔAUROC null even with diverse critics $\rightarrow$ "beyond-score critic uncertainty carries no silent-failure information — agent safeguards built on it inherit the critic's blind spots." H2 flat $\rightarrow$ diversity doesn't rescue detection. H1 holds, H3 fails $\rightarrow$ uncertainty helps only under mild selection; report the operating envelope.

4 Formal Problem Definition and Estimand

Setup. A frozen agent π runs on tasks $x \in \mathcal{X}$ from verifiable benchmarks. At each decision point it produces candidate actions (or whole trajectories) scored by a critic c (a process reward model, generative verifier, or LLM judge). Two selection constructions, both reported: (a) *fixed-percent* — the top $k\%$ of candidates by c; (b) *taken / best-of-n* — the action the agent actually commits to over draws of size n, the deployment-relevant set, with pressure KL $\approx \log n - (n-1)/n$.

Gold. $g \in \{0, 1\}$: at the *trajectory* level, task success (tests pass / environment reward) — clean and free on verifiable benchmarks. At the *step* level, "the step lies on a still-succeeding path," estimated by Monte-Carlo rollouts from that state (the Math-Shepherd construction), reported with a rollout-noise sensitivity analysis.

Estimand (primary). The incremental discrimination of uncertainty u over critic score c, conditional on selection:

$$\underbrace{\Pr\big(g{=}0 \mid c,\ \text{sel}\big)}_{\text{base}} \quad \text{vs.} \quad \underbrace{\Pr\big(g{=}0 \mid c,\ u,\ \text{sel}\big)}_{\text{full}}, \qquad \Delta\text{AUROC} = \text{AUROC}(c, u) - \text{AUROC}(c).$$

Report ΔAUROC, partial AUC (low-alarm-rate regime), and the likelihood-ratio test for u, with inference clustered by task/trajectory. The selection-induced correlation between c and u is reported as a range-restriction diagnostic.

Uncertainty u. Primary: disagreement across a critic ensemble $\{c_1 \dots c_m\}$ (std / range / rank disagreement). Cheap: dropout, quantile-width, generative-verifier self-consistency. Trivial covariates always in the model: critic score, action length, step index, and the agent's own token entropy.

Critic diversity (manipulated). seed → data → base model → public-critic zoo, with diversity *measured* (pairwise error correlation on gold, on the full candidate set and the selected set; prediction disagreement; CKA).

5 Design, Statistical Plan, and Experiments

Fractional design. An anchor — base-diverse critics, m=4, ensemble-std, best-of-n=16, k=5%, SWE-bench-Verified gold, one open agent scaffold — plus one-factor-at-a-time sweeps. **Preregistration:** primary endpoint is ΔAUROC at the anchor, diverse vs seed critics, clustered-bootstrap 95% CI; other contrasts under Benjamini–Hochberg FDR. **Confounds:** action length, step index, agent entropy always in the conditional model; range-restriction diagnostic; rollout-noise sensitivity for step-level gold.

ID	Question	Design	Metrics	Tier
E1	Beyond-score detection (H1)	anchor ± selection sweep (n, k, taken set)	ΔAUROC, partial AUC, LR vs n, k	Core
E2	Critic diversity + mediation (H2)	diversity sweep × m at anchor	ΔAUROC by diversity; error-corr. mediation	Core
E3	Pressure × diversity (H3)	seed vs base-diverse × n grid	slope of ΔAUROC in $\log n$	Core
E4	Cheap uncertainty (H4)	estimator sweep at anchor	incremental ΔAUROC per estimator	Ablation
E5	Failure taxonomy (H5)	labeled failures across classes	per-class ΔAUROC; per-class error corr.	Mechanism
E6	Generality	anchor across SWE-bench / WebArena / ALFWorld	cross-benchmark agreement	Robustness
E7	Backtrack / selective verify (offline)	best cells; logged-trajectory simulation	failures caught at b% budget; risk–coverage	Impact
E8	Trajectory- vs step-level gold	task-success vs rollout labels	agreement; step-level ΔAUROC	Scope

6 Doubts / Risks

- **Step-level gold is the hard part.** "Was this step a mistake" is not directly verifiable; the MC-rollout label (Math-Shepherd style) is an approximation — so trajectory-level gold (task success) is the clean primary, step-level the flagged extension, with rollout-noise sensitivity.
- **The null is publishable and safety-relevant.** If uncertainty adds nothing beyond the critic score even with diverse critics, agent safeguards built on it inherit the critic's blind spots — a result worth stating.

- **Range restriction is the crux, not a nuisance.** Selection ties c and u together in the selected set; the estimand controls for c by construction and the induced correlation is reported directly.
- **Critic panels are heterogeneous.** Different critics use different formats/scales; scores are rank-normalized before measuring disagreement, and per-critic quality (RewardBench-style) is reported.
- **Offline scope for E7.** Adaptive backtrack-then-retry creates selection feedback a fixed log cannot simulate; policy simulation is offline in v1, with the caveat in limitations.

7 Benchmarks

- **SWE-bench / SWE-bench Verified** (arXiv:2310.06770) — primary: coding agents, gold = unit tests pass; clean, objective trajectory-level labels.
- **WebArena** (arXiv:2307.13854) — long-horizon web tasks with checkable success; the goal-drift / doom-loop regime.
- **ALFWorld** (arXiv:2010.03768) — embodied/text household tasks with programmatic success; a third, distinct domain for generality (E6).
- **Critics**: process reward models (arXiv:2305.20050; arXiv:2312.08935), generative verifiers (arXiv:2408.15240), agent PRMs (arXiv:2511.08325), and public reward models from Hugging Face for the maximal-diversity zoo. Per-critic quality anchored to RewardBench (arXiv:2403.13787).

Metrics: ΔAUROC and partial AUC (primary), LR test, unconditional AUROC (secondary), per-class ΔAUROC, error-correlation mediation, and — for E7 — silent failures caught at a b% backtrack/verification budget with risk–coverage.

8 Positioning

Work	Selection-focused	Task-verified	Uncertainty as detector	Beyond-score	Critic diversity manipulated	Failure taxonomy
PRMs (arXiv:2305.20050; arXiv:2312.08935)	partial	✓	× (scoring)	×	×	×
Generative Verifiers (arXiv:2408.15240)	partial	✓	× (scoring)	×	×	×
AgentPRM (arXiv:2511.08325)	✓	✓	× (scoring)	×	×	×
Reflexion (arXiv:2303.11366)	×	✓	× (repair)	×	×	×
Agent failure detection (arXiv:2605.08715; arXiv:2608.02464)	✓	✓	✓	×	×	partial
URM/URME (arXiv:2410.00847)	×	× (BoN qual.)	✓ uncond.	×	× (seed)	×
This work	✓	✓	✓	✓	✓ 4-lvl, mediated	✓

Open cell: selection-conditional, beyond-score silent-failure detection in agents, task-verified, with critic diversity manipulated and mediated.

9 Value / Feasibility

Runs on frozen open agents and off-the-shelf / lightly-trained critics: log trajectories once on open benchmarks, score with a critic panel, label gold from task success, then the whole study is offline

scoring plus regression — *no RL*, single node. The public-critic zoo is essentially free (download + inference). The payoff is a calibrated silent-failure trigger for backtracking / selective verification — directly the reliable-agent loop — and a mechanistic account (diversity → decorrelation) of when it works. **Staged:** Stage 0 (vacuousness/power: do silent failures exist in the selected set? is the failure rate workable? do trivial features saturate?) → Stage 1 (pre-registered anchor + diversity ladder; a signal promotes, a flat result is a negative-result paper) → Stage 2 (full matrix E1–E8, three benchmarks, taxonomy, step-level gold).

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